INCH-POUND

A-A-54040
19 March 1993

COMMERCIAL ITEM DESCRIPTION

REGULATOR, OXYGEN PRESSURE

(Lightweight with Integral Flow Controller)

The General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description covers an oxygen regulator with two 50 psig outlet ports and an integral flow controller.

Salient characteristics:

The regulator shall be an oxygen pressure regulator with an oxygen flow controller for medical use. It shall be suitable for the operation of two demand valves or one demand valve and an aspirator. It shall consist of:

* regulator
* flow controller
* gauge for supply pressure
* yoke for oxygen cylinder valve

The unit shall have a total of three external outlet ports, two factory set 50 psig ports with, at least , 100 LPM flow rate and one 2 LPM to 15 LPM flow controller port. When supplied with cylinder pressures between 500 and 2000 psig pressure, all three outlet ports shall be usable simultaneously. The 50 psig outlet ports shall maintain 50 psig plus or minus 10 psig and a minimum flow of 100 LPM at each 50 psig port.

AMSC N/A FSC 6680

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

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The envelope size of the regulator shall not exceed 4 inches by 4 inches by 6 inches. The supply oxygen shall be internally filtered, stopping all particles larger than 25 microns.

In addition to requirements herein, the regulator and components shall be in accordance with CGA E-7. Also, any materials of construction in contact with oxygen shall be resistant to the action of oxygen and these materials shall be in conformance with requirements of UL 252.

The regulator shall provide an internal port to supply oxygen to the flow controller. The flow controller outlet port shall be CGA, V-5, No. 1240 barbed end to accept 1/4 inch ID tubing. The flow controller shall have a selectable flow control which is marked to provide 2, 4, 6, 10, and 15 LPM flows. There shall be zero (0) leakage at the flow controller outlet port when it is set to zero/off position and the regulator is supplied with 2000 psig.

The two 50 psig outlet ports shall be DISS, V-5, No. 1240, 9/16 inch male threads.

Either single stage or two stage pressure reduction is acceptable and either piston or diaphragm design is acceptable.

The regulator shall have a safety relief valve which shall prevent outlet pressure in excess of 100 psig in the event of regulator failure. A safety relief valve which operates at an outlet pressure above the set outlet pressure and below 100 psig is acceptable.

The gauge shall be 1-1/2 inch diameter with a rubber bumper to protect it from shock. The gauge shall indicate the cylinder pressure and be calibrated in psig and Kg/cm^2 (or Kpa). The gauge shall be marked to indicate a full cylinder tank at tank pressure of approximately 1900 psig. In addition to CGA E-7 requirement that gauges meet the requirements of ANSI B40.1, the gauge shall have a minimum accuracy of grade B as defined in ANSI B40.1, shall have a range of 0-4000 PSIG, and shall be in accordance with UL 404.

The yoke shall be integral with the regulator and be pin-indexed for size D and E medical oxygen cylinder valve in accordance with CGA, V-1, No. 870.

The regulator, with all components, shall be packaged in a closed plastic bag to prohibit dust collection.

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Two owner's manuals shall be included providing, at least, information required by CGA E-7.

The complete unit shall conform to applicable requirements of a nationally recognized standard for safety. The complete unit shall be approved as conforming to such a standard by an independent nationally recognized testing laboratory (i.e. UL listed or approved by CSA, ETL, or City of Los Angeles).

Workmanship. The regulator shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One regulator, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 1.0 (percent defective) and an inspection level of II. The unit of product for examination purposes shall be one regulator, as specified in this description.

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Sampling for tests. Sampling for tests, including dimensional test, shall be conducted in accordance with MIL-STD-105, with an AQL of 1.0 (percent defective) and an inspection level of II. The unit of product for test purposes shall be one regulator.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics. Tests shall include, but shall not be limited to, the following:

a. Minimum pressure and flow rate and maximum pressure at regulator 50 psig outlet ports. Provide an oxygen source of 500 psig to the yoke inlet. Set the flow controller at 15 LPM. Measure the pressure at both 50 psig outlet ports (required to be between 40 and 60 psig). Measure the flow rate at both 50 psig outlet ports simultaneously (required to be at least 100 LPM).

b. Using standard laboratory testing equipment, with an oxygen supply source of 1900 to 2000 psig and the two 50 psig outlet ports occluded, measure the flow rate at the flow controller outlet port five times, once for each setting of 2, 4, 6, 10, and 15 LPM (required to be within 25 percent of marked setting).

c. Test for zero leakage at the flow controller outlet port with supply pressure to the regulator of 1900 and 2000 psig.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

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Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document.

6680-01-234-6789

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MILITARY INTERESTS:

Custodians:

Air Force - 03
Army - MD
Navy - MS

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

FDA-MPQAS
USPHS
VA - OSS

Project Number: 6680-0216

Location: REGUL12C.DOC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER A-A-54040	2. DOCUMENT DATE (YYMMDD) 19 March 1993

3. DOCUMENT TITLE
REGULATOR, OXYGEN PRESSURE

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME DEFENSE PERSONNEL SUPPORT CENTER ATTN: DPSC-MSE	b. TELEPHONE (Include Area Code) (1) Commercial: AC 215-737-2870 (2) AUTOVON: AV 444-2870
c. ADDRESS (Include Zip Code) 2800 SO. 20TH STREET PHILADELPHIA, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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